

Fig. 2: An actual-size, PCB layout of the dual audio-signal tracer

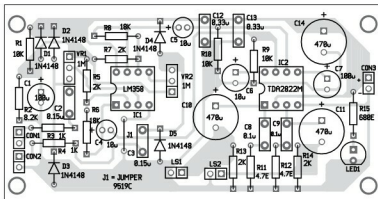


Fig. 3: Component layout of the PCB

channel, respectively. Op-amps of IC1 are configured as non-inverting amplifiers with gain fixed at ten. Voltage gains can have any values starting from one, depending on the need. However, it is not a good idea to have voltage gains higher than 30 because then the circuit becomes too sensitive to electromagnetic fields and may produce a lot of noise in the loudspeakers.

IC2 works as a dual-channel output amplifier. Each amplifier works with internally-fixed gain of around 100. The amplifier can drive loudspeakers and

headphones with resistance equal to or higher than 8-ohm.

Construction and testing

An actual-size, single-side PCB of the dual audio-signal tracer is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic box.

The circuit works well off 6V battery; the preferred power supply is 6V and above. The circuit is appropriate for the signal tracer, as

general-purpose dual audio amplifier and as low-power stereo amplifier. The output power can reach 1W over 4-ohm with 9V power supply, which is more than enough for listening to the produced sound in a small room. ●

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